

Phases and Paths of the Spectral-Entropy Universality Programme

thesis-IV working notes (HyMeKo)

2026-04-26

Abstract

This document is a single-source reference for the spectral-entropy regulariser programme. It defines every *phase* (the experimental campaigns run in chronological order) and every *path* (the regulariser variants tested), gives the configuration of each, and records the empirical headline of each. It is intended for paper preparation and future re-runs — not for results interpretation (see `RESULTS_VIEWS_SUITE.md` and the per-phase reports for that).

1 Overview

The programme tests whether a spectral-entropy regulariser inspired by Hajdu thesis IV can be made *universal* across architectures and datasets. Eight phases were run between 2026-04-22 and 2026-04-26. Twelve regulariser *arms* (paths) were defined and benchmarked against a paired **baseline** (no regulariser). All comparisons are paired by seed to share weight initialisation and DataLoader shuffle.

Total experiments to date. 98+ paired runs $\times$ $\sim$ 2–300 seeds = roughly 11 000 individual training runs.

Four anchors.

- **MNIST plain MLP:** original $\Delta = +0.149$ pp positive (the seed of the programme).
- **Spirals 2-D synthetic:** largest positive ($+0.412 \rightarrow +0.724$ pp across paths).
- **Circles 2-D synthetic:** original -0.060 pp significant negative (stress test).
- **CapsMLP MNIST:** original -0.057 pp significant negative (stress test).

2 Datasets and architectures

2.1 Datasets

Identifier	Origin	Train size	Classes	Notes
<code>mnist</code> , <code>mnist_small</code>	MNIST	60k	10	28 $\times$ 28 grayscale
<code>fashion_mnist</code>	FashionMNIST	60k	10	28 $\times$ 28 grayscale (clothing)
<code>kmnist</code>	KMNIST	60k	10	28 $\times$ 28 grayscale (Japanese hiragana). HF tanganke/kmnist mirror; official torchvision URL is dead.
<code>emnist_letters</code>	EMNIST	145k	26	28 $\times$ 28 grayscale (letters)
<code>cifar10</code>	CIFAR-10	50k	10	32 $\times$ 32 colour
<code>svhn</code>	SVHN	73k	10	32 $\times$ 32 colour
<code>iris</code> , <code>wine</code> , <code>breast_cancer</code> , <code>digits</code>	sklearn	100–500	2–10	tabular
<code>two_moons</code> , <code>spirals</code> , <code>circles</code> , <code>gaussian_quants</code>	synthetic	2000	2–3	2-D classification

2.2 Architectures

Identifier	Class	Description
MNISTNet	plain MLP	784→256→128→64→10
MNISTNetSmall	plain MLP, thesis	784→16→8→10 (the anchor)
MNISTNetSmallNClass	plain MLP, parametric	784→16→8→ n (used for EMNIST)
ResMLP	residual MLP	projection + N residual blocks at hidden 16
HighwayMLP	gated MLP	projection + N highway blocks at hidden 16
CIFAR10Net	MLP	3072→64→32→10 (proj. skipped from regulariser)
CIFAR10ResMLP	residual MLP	projection 3072→16, then N residual blocks
SVHNNet	MLP	3072→64→32→10 (similar to CIFAR10Net)
TabularMLP	plain MLP, parametric	input→32→16→ n (sklearn datasets)
SyntheticMLP	plain MLP, 2-D	2→32→16→2 (two_moons, spirals, circles)
SyntheticMLP3	plain MLP, 2-D	2→32→16→3 (gaussian_quants)
CapsMLP	capsule network	flat 784 → 32 primary caps of 8-D → 10 digit caps of 16-D, dynamic routing (3 iters), squash activation

3 Regulariser arms (paths)

All arms share the optimisation outer loop: Adam, λ weight on the regulariser, every **reg-every-n** batches the regulariser is recomputed from `model.spectral_adjacency(view)` and its (data-dependent or weight-only) statistics.

3.1 Originals (phases 1–5)

baseline

No regulariser; `reg_term` = 0. Always paired against every treated arm.

scalar_entropy

$\mathcal{L}_{\text{reg}} = \lambda \cdot H(A)$, where H is Shannon entropy of normalised-Laplacian eigenvalues of the chosen-view adjacency. $\lambda=0.1$ default.

kl_trajectory

$\mathcal{L}_{\text{reg}} = \lambda \cdot D_{KL}(\hat{\lambda}^{(t-1)} \parallel \hat{\lambda}^{(t)})$. Penalises step-to-step divergence of the spectrum.

l2_weight_decay

Standard L^2 control arm: $\mathcal{L}_{\text{reg}} = \lambda \sum_l \|W_l\|_F^2$.

3.2 Universality candidates (phases 6–7b)

scalar_entropy_normalized (*Path B*)

$\mathcal{L}_{\text{reg}} = \lambda \cdot H(A) / \log_2 \text{rank}(A)$. **Scale-invariant**; ties to VC / Rademacher effective-rank bounds.

entropy_target (*Path A*)

$\mathcal{L}_{\text{reg}} = \lambda \cdot (H_{\text{norm}} - H^*)^2$ with $H^* \in [0, 1]$. Squared-deviation around an information-bottleneck-style compression target. $H^* = 0.5$ is the universal-default sweet spot.

structural_composite (*Path C*)

$\mathcal{L}_{\text{reg}} = \lambda \cdot \max(H_{\text{norm}}, \sigma_{\text{max}}/2, 1 - \text{sr}/\text{rank})$. Penalises whichever of three Bartlett-Foster-Telgarsky bound terms is largest at the moment.

entropy_unified

$\mathcal{L}_{\text{reg}} = \lambda \cdot ((H_{\text{norm}} - H^*)^2 + \beta H_{\text{norm}})$. Convex blend of A and B; $\beta=1.0$ default makes it numerically equivalent to $H^*=0.0$ entropy_target.

entropy_adaptive

$\mathcal{L}_{\text{reg}} = \lambda(t) \cdot H_{\text{norm}}$, with $\lambda(t) = \lambda \cdot \exp(\eta \cdot \ddot{H}_{\text{norm}})$ clamped to $[0.1, 10]$. Multiplicative modulator on λ driven by the *second derivative* of H_{norm} across reg-update steps.

total_combined

$\mathcal{L}_{\text{reg}} = \lambda \cdot (A + B + C)$. Literal sum of the three; triple-counts H_{norm} . Robustness-oriented.

entropy_target_ka

$\mathcal{L}_{\text{reg}} = \lambda \cdot (H_{\text{norm}} - H_{KA}^*)^2$, with $H_{KA}^* = \max(0.5, \min(1.0, \log_2(2n+1)/\log_2(\text{rank})))$. **Theory-driven default:** the Hecht-Nielsen / Kolmogorov-Arnold representation theorem gives a per-dataset lower bound on the minimum representational rank ($2n+1$ for n -dim input).

entropy_telgarsky

Per-layer entropy with depth-decay weighting: $\mathcal{L}_{\text{reg}} = \lambda \sum_l (1-l/L)^p H_{\text{norm}}^{(l)}$. Operationalises Telgarsky’s depth-vs-width tradeoff: deeper layers carry less penalty (preserve information late, compress early).

3.3 Activation-side family (phases 8 & 11)

cross_layer_mi (*Path F*)

$\mathcal{L}_{\text{reg}} = \lambda \cdot \frac{1}{\binom{L}{2}} \sum_{i < j} I_2(\hat{K}_i; \hat{K}_j)$, where $\hat{K}_l$ is the trace-1 Gaussian-RBF Gram matrix of layer l ’s post-linear activations on the current mini-batch and I_2 is the Sanchez-Giraldo Rényi-2 mutual information via the Hadamard-product joint. Penalises layer redundancy; differentiable through activations to weights via forward hooks. Math: `reports/sanchez-giraldo-framework.{md,pdf}`.

3.4 Multi-information + closed-loop control (phases 12–18, Path I)

total_correlation_mi (*Path I*)

Strict L-way generalisation of Path F: $\text{TC}_2 = \sum_{l=1}^L H_2(\hat{K}_l) - H_2(\hat{K}_{\text{join}})$ where $\hat{K}_{\text{join}} = \bigodot_l \hat{K}_l / \text{tr}(\bigodot_l \hat{K}_l)$ is the stepwise-renormalised Hadamard joint Gram. Wrapped with two closed-loop modulators: (i) Lyapunov-style λ schedule $\lambda_{\text{eff}} = \lambda_0 \exp(-\eta \text{KL}_{\text{step}})$ where KL_{step} is the eigenvalue-distribution KL between consecutive joint-Gram spectra; (ii) variance-momentum modulator $\mu_t = \beta \mu_{t-1} + (1-\beta) \sigma^2(\hat{p}_t)$ with three CLI-selectable mixing modes (`damp`, `amplify`, `mix`). Math + design discussion: `reports/ph12_path_i_negative_result.md`.

entropy_lyapunov (*multi-term wrapping arm*)

Wraps Path A + Path B with the same Lyapunov KL-feedback schedule, parameterised by a per-term weight pair (λ_a, λ_b) : $\mathcal{L}_{\text{reg}} = \lambda_{\text{eff}} (\lambda_a (H_{\text{norm}} - H^*)^2 + \lambda_b H_{\text{norm}})$. The KL signal in the schedule connects two previously independent regularisers (`kl_trajectory` and `entropy_target`) into a single closed-loop controller derived from $\dot{V} < 0$ on the task loss.

3.5 Generated nets via HyMeKo descriptions (phases 18, 18b)

The Tier-2 emission path of the `torch_dataflow` backend (see `transforms/torch_dataflow/template.py`) is exercised end-to-end as a regularisation-bench fixture: `.hymeko` sources in `data/nn/` for ResMLP- $\{3,10,20,40\}$ and HighwayMLP- $\{3,10,20\}$ compile to PyTorch modules backed by `ehk_torch_stub` blocks, register as `\_hymeko`-suffixed datasets in `run_benchmark.py`, and run the same regulariser arms as the hand-written PyTorch counterparts. Confirms that the canonical IR round-trips through to runnable nets without distorting regulariser dynamics (architectural-parity claim).

4 Phases

Each phase is one overnight (or multi-overnight) campaign with its own shell script and log file. Phases share the same harness (`run_benchmark.py`) and CSV output schema (`data/benchmarks/thesis_iv_hard_*.csv`).

Phase 1 — Original overnight suite (2026-04-22 – 23)

Scripts: `run_overnight.sh`, `run_overnight_views.sh` (2nd suite). **Goal:** establish whether the unnormalised spectral entropy regulariser yields a positive on MNIST plain MLP and whether the effect generalises to image siblings, deep architectures, and CIFAR-10. **Arms:** `scalar_entropy`, `kl_trajectory`. **Datasets / archs:** MNIST, FashionMNIST, KMNIST (failed at the mirror), CIFAR-10, ResMLP-20, HighwayMLP, plain MLPs. **Headline:** MNIST anchor confirmed at +0.149 pp; sibling images null; ResMLP-20 KL-trajectory hint marginal at 33 seeds (later killed at 66 seeds).

Phase 4 — Highway + tabular (2026-04-23)

Script: `run_overnight_views_ph4.sh`. **Goal:** extend coverage to tabular sklearn datasets and the HighwayMLP/ResMLP-20 depth sweep. **Arms:** `scalar_entropy`, `kl_trajectory`. **Datasets / archs:** iris, wine, breast_cancer, digits, MNIST HighwayMLP-10/20, FashionMNIST HighwayMLP-20, FashionMNIST ResMLP-20. **Headline:** digits significant positive (+0.047 pp, $p < 0.01$); rest mostly null due to ceiling effects.

Phase 5 — New datasets and CapsMLP (2026-04-23 – 24)

Script: `run_overnight_views_ph5.sh`. **Goal:** test the original arm on Gaussian-quantiles synthetic, EMNIST Letters, SVHN, and the new CapsMLP architecture. **Arms:** `scalar_entropy`, `kl_trajectory`. **Headline:**

CapsMLP MNIST surfaced as a *significant negative* (-0.057 pp, $p < 0.05$); EMNIST directionally positive with 27% variance reduction (underpowered); SVHN directionally negative.

Phase 6 — Universality stress test (2026-04-25)

Script: `run_overnight_views_ph6.sh`. **Goal:** test whether Path B / Path A / Path C neutralise the four significant negatives surfaced in phases 1–5. **Arms:** `scalar_entropy_normalized`, `entropy_target` ($H^*=0.5$), `structural_composite`. **Datasets / archs:** spirals, circles, mnist_small, mnist_capsnet (all dataflow view). **Headline:** Path B and Path A both *strictly Pareto-dominate* `scalar_entropy` on the 4-dataset matrix — amplify positives (MNIST $+0.232$, spirals $+0.624/+0.662$) and neutralise both negatives (CapsMLP -0.002 , circles 0.000 exactly).

Phase 7 — H^* sweep + combined arms (2026-04-25)

Script: `run_overnight_views_ph7.sh`. **Goal:** (a) map the H^* curve at $\{0.0, 0.25, 0.5, 0.75, 1.0\}$ on synthetic; (b) test whether combined arms (`entropy_unified`, `entropy_adaptive`, `total_combined`) outperform pure A or B. **Headline:** H^* peaks at 0.5 (circles) and 0.75 (spirals); $H^*=1.0$ is degenerate (asks for max entropy — collapses). `entropy_adaptive` sets a new spirals record ($+0.724$ pp) but is mid-pack on MNIST. `total_combined` achieves the highest seed-level win-rate on MNIST (29/4) at the cost of mean magnitude.

Phase 7b — KA + Telgarsky (2026-04-25 – 26)

Script: `run_overnight_views_ph7b.sh`. **Goal:** test theory-driven defaults — a per-dataset H^* derived from the Kolmogorov-Arnold representation theorem (`entropy_target_ka`) and Telgarsky-style per-layer depth-weighted entropy (`entropy_telgarsky`). **Headline:** `entropy_target_ka` reproduces fixed $H^*=0.5$ exactly on synthetic (theory-data agreement $\checkmark$). MNIST H^*_{KA} clamps at 1.0 due to ceiling; implementation bug — ceiling should be 0.75. `entropy_telgarsky` positive on spirals ($+0.598$ pp) but *significantly negative on circles* (-0.046 pp, $p < 0.05$). Telgarsky depth-weighting fails universality.

Phase 7c — Universality breadth (queued, 2026-04-26)

Script: `run_overnight_views_ph7c.sh`. **Brief:** `reports/phase7c_brief.\{md,pdf\}`. **Goal:** extend Path B and Path A to six previously-unseen dataset/arch combos (ResMLP-20, FashionMNIST, KMNIST, SVHN, EMNIST, HighwayMLP-20). Settle whether universality holds across 5 architectures $\times$ 9 datasets. **Status:** auto-fires after 7b.

Phase 8 — Cross-layer MI (queued, 2026-04-26)

Script: `run_overnight_views_ph8.sh`. **Math:** `reports/sanchez_giraldo_framework.\{md,pdf\}`. **Goal:** test the activation-side Sanchez-Giraldo Rényi-2 cross-layer mutual information regulariser. Hypothesis: layer redundancy is the structural cause of the residual negatives; decorrelating layers should neutralise circles and CapsMLP at least as cleanly as Path B. **Status:** auto-fires after 7c.

Phase 9 — Composability test (queued, 2026-04-26)

Script: `run_overnight_views_ph9.sh`. **Brief:** `reports/phase9_brief.\{md,pdf\}`. **Goal:** test whether `scalar_entropy_normalized` composes with the established mainstream regularisers (dropout, batch norm, optimiser weight decay). Five model+optimiser variants (`vanilla`, `dropout`, `batch_norm`, `weight_decay`, `full_stack`) tested across the 4-dataset stress matrix; paired baseline vs `scalar_entropy_normalized` within each variant. Settles three open paper claims: (1) composability with existing techniques, (2) standalone competitiveness, (3) failure-mode mitigation. **Status:** 12 new DATASETS variants registered (`*_drop`, `*_bn`, `*_full`); new `--weight-decay` CLI flag plumbed through. Auto-fires after 8.

Phase 10 — Lyapunov closed-loop scaffold (2026-04-26)

Script: `run_overnight_views_ph10.sh`. **Goal:** stress-test the new `entropy_lyapunov` arm in the multi-task / multi-dataset matrix (per-term λ_a, λ_b plus KL-feedback schedule). Acts as the deployment scaffold that phases 12 onwards extend. **Headline:** closed-loop schedule trains stably across the matrix; the multi-term arm becomes the workhorse for phase 16 (wrapping Path A/B) and phase 18 (deep architectures).

Phase 11 — Path F characterisation (2026-04-26)

Script: `run_overnight_views_ph11.sh`. **Goal:** characterise `cross_layer_mi` (Path F) across λ and across datasets following the 2026-04-26 spirals null ($\Delta = +0.0003$, $t = +0.16$, $W/L = 28/54$ at $\lambda = 0.1$). **Result:** 9 paired runs, all null on plain MLPs; spirals λ sweep flat; cross-dataset (circles, `mnist_small`, `fashion_mnist`, `kmnist`) null at all tested operating points. **Path F is honestly weaker than Path A/B at the conventional operating point** — the activation-side pairwise MI does not substitute for normalised spectral entropy.

Phase 12 — Path I sweep (2026-04-26 – 27)

Script: `run_overnight_views_ph12.sh`. **Brief / negative-result writeup:** `reports/ph12_path_i_negative_result.md`. **Goal:** characterise `total_correlation_mi` (Path I) — the L-way joint multi-information generalisation of Path F with KL feedback and variance momentum — across (a) variance modes (`damp` / `amplify` / `mix`), (b) λ sweep on spirals, (c) cross-dataset matrix, (d) momentum β sweep. **Headline (13 runs):** at default $\lambda = 0.1$ Path I is *significantly worse than baseline* on spirals ($\Delta = -1.014$ pp, $t = -4.62$, $W/L = 8/78$, $p < 0.001$) under `mix`; the same magnitude under `amplify`; `damp` mode is degenerate (regulariser fades to zero). λ sweep maps a catastrophic-collapse curve: $\Delta = -26.572$ pp at $\lambda = 1.0$. **Sole positive trend:** $\lambda = 0.03$ on spirals at $\Delta = +0.306$ pp, $t = +1.78$, $p \approx 0.08$ — borderline significant. Cross-dataset at $\lambda = 0.01$ all null; high- β EMA momentum ($\beta = 0.99$) mildly attenuates the negative.

Phase 13 — Path I $\times$ CapsMLP (2026-04-27, INVALIDATED)

Script: `run_overnight_views_ph13.sh`. **Result:** 5/5 paired runs with $\Delta = +0.000$ exactly. Root cause: `CapsMLP.spectral_weights()` returns `[primary.weight, w_matrix]` where `w_matrix` is a *synthesised* tensor (`self.W.permute().reshape()`) not backed by an `nn.Linear`; the activation-hook setup `id(m.weight)` in `weight_ids` consequently finds only one hookable module, the `len(activations) >= 2` guard fails, and `reg_term` stays `None` for every batch — training is byte-identical to baseline. **Fix landed:** the activation-hook setup now emits a [WARN] line whenever fewer than two hookable modules are found, so the silent-collapse failure mode is eliminated going forward. Phase 13 is informationally vacuous about CapsMLP and must be re-run after CapsMLP is refactored to expose its routing as a hookable submodule.

Phase 14 — Path I $\times$ deep architectures (2026-04-27)

Script: `run_overnight_views_ph14.sh`. **Goal:** test whether the L-way joint Gram bites at $L=20\text{--}40$ where the pairwise Path F could not. Datasets: ResMLP-20, HighwayMLP {10, 20} on MNIST; ResMLP-20 on FashionMNIST. **Headline (5 runs, scouting power 15 seeds):** all five null; `mnist_highway_20` the strongest signed effect at $\Delta = +0.057$ pp, $t = +0.99$ — below the +0.10 acceptance threshold. The deep-arch hypothesis does not pan out at scouting power.

Phase 15 — Path I fine-grained λ sweep (2026-04-27)

Script: `run_overnight_views_ph15.sh`. **Goal:** resolve the $\lambda = 0.03$ spirals positive at finer resolution: $\lambda \in \{0.005, 0.010, 0.020, 0.030, 0.050\}$, `mix` mode. **Headline:** $\lambda = 0.03$ peak *reproduces to three decimals* ($\Delta = +0.306$ pp, $t = +1.78$, $W/L = 40/31$, $p \approx 0.08$ again). Productive band $[0.005, 0.05]$ is uniformly positive but not significant at the other points; flat-topped maximum around $\lambda = 0.02\text{--}0.03$.

Phase 16 — Multi-term wrapping on Path A/B (2026-04-27)

Script: `run_overnight_views_ph16.sh`. **Goal:** test whether the Lyapunov KL-feedback schedule helps when wrapped around the universality *winners* (Path A, Path B) rather than around Path I. Three datasets (spirals, circles, `mnist_small`) $\times$ `baseline` + Path B + `entropy_lyapunov` with $(\lambda_a, \lambda_b) \in \{(0, 1), (1, 0)\}$. **Headline:** multi-term wrapping does not lift Path B beyond its plain version on these fixtures — the schedule is mechanically fine but does not visibly contribute when the base regulariser is already universal.

Phase 17 — Path I λ/L rescaling + more iterations (2026-04-28)

Script: `run_overnight_views_ph17.sh`. **Goal:** apply the calibration law $\lambda_I \approx \lambda_B/L$ on deep architectures, and triple the epoch budget to 30. Datasets: `mnist_highway_10` ($\lambda = 0.001$), `mnist_highway_20` ($\lambda = 0.0005$), `mnist_resnet_40` ($\lambda = 0.0001$). **Headline (3 runs, 15 seeds):** `mnist_highway_10` null ($\Delta = +0.001$); `mnist_highway_20` marginal positive trend ($\Delta = +0.073$ pp, $t = +1.65$, $p \approx 0.10$); `mnist_resnet_40` similar weak null. Rescaled λ does *not* rescue Path I to a publishable lift.

Phase 18 — Multi-term $\times$ deep $\times$ HyMeKo descriptions (2026-04-28 – 29)

Scripts: `run_overnight_views_ph18.sh`, `run_overnight_views_ph18b.sh`. **Goal:** two combined buckets: (A) PyTorch hand-written deep archs $\times$ FashionMNIST under `entropy_lyapunov` ($\lambda_a = \lambda_b = 1$, $\eta = 5$), and (B) the same arm against six HyMeKo-emitted nets (`mnist\_highway\_3,10,20\_hymeko`, `mnist\_resmlp\_10,20,40\_hymeko`) compiled from `data/nn/*.hymeko` via the `torch\_dataflow` backend. **Headline:** `fashion_mnist_highway_10` significant positive ($\Delta = +0.087$ pp, $t = +2.61$, $p < 0.01$) under multi-term wrapping — the only published-power positive of the multi-term programme so far; deeper variants null or trend negative. HyMeKo-generated bucket B *statistically mirrors* bucket A (highway-20 directional positive, resmlp-40 directional negative), confirming the architectural-parity claim end-to-end. Bucket B required a small wrapper fix in `run_benchmark.py::\_wrap\_hymeko\_net` (flatten 28×28 inputs to flat 784) — shipped as `ph18b`.

Phase 18c — Confirmation sweep + depth λ -grid (2026-05-04)

Script: `run_overnight_views_ph18c.sh`. **Goal:** two follow-on questions raised by `ph18`: (1) does the `fashion_mnist_highway_10` positive survive a 10-seed sweep? (2) is the depth-fragility of highway-20 / resmlp-40 real, or was the original constant- λ off-axis at depth? Path I’s calibration law $\lambda_{\text{multi}} \sim \lambda_{\text{scalar}}/L$ predicts the original sweep was systematically over-strong at depth. The cell grid is highway-10 confirm 10-seed $\times \lambda=1.0$, plus highway-20 / resmlp-40 / resnet-20 each at $\lambda \in \{0.5, 2.0, 5.0\}$ (3-seed), plus a highway-20 10-seed cross-check at $\lambda=1.0$. **Findings:**

- *Headline confirmed.* `fashion_mnist_highway_10` 10-seed: $\Delta = +0.00092$ (+0.092 pp), $t = +2.08$, $p < 0.05$, W/L/T = 9/1/0. The load-bearing positive of the multi-term programme survives the bedrock test.
- *New positive at depth.* `mnist_resmlp_40_hymeko` at $\lambda_a=\lambda_b=0.5$: $\Delta = +0.00213$ (+0.213 pp), $t = +2.72$, W/L/T = 3/0/0 — significant positive at $L=40$ once λ is rescaled. This is the first empirical validation of Path I’s calibration law on the multi-term arm: the original `ph18` null at $\lambda=1.0$ was a depth-mistuning artefact for resmlp-40, not a depth-fragility limit.
- *Depth-fragility confirmed for highway-20.* The 10-seed cross-check at $\lambda=1.0$ returns $\Delta = -0.00022$ ($t = -0.35$, W/L/T = 4/6/0); the full $\lambda \in \{0.5, 1.0, 2.0, 5.0\}$ grid all return null or trending-negative. Highway-20’s depth-fragility is therefore not a λ -tuning artefact.

Outcome. The multi-term programme has *two* confirmed positives (highway-10 at $L=10$ and resmlp-40 at $L=40$ with depth-aware λ) plus one genuine null (highway-20). Together with `ph12` / `ph15`, this gives Path I’s calibration law two independent empirical anchors: the spirals collapse at $\lambda \geq 0.1$ at $L=4$, and the resmlp-40 recovery at $\lambda=0.5$ at $L=40$.

5 Headline results map

Per-arm best-known result on each anchor dataset (most recent published-power configuration; see `RESULTS_VIEWS_SUITE.md` for the full table).

arm	spirals (+)	MNIST (+)	circles (–)
<code>scalar_entropy</code>	+0.412 ***	+0.149 **	
<code>kl_trajectory</code>	+0.294 ***	marginal	
<code>scalar_entropy_normalized</code> (B)	+0.624 ***	+0.232 ***	
<code>entropy_target</code> $H^*=0.5$ (A)	+0.662 ***	+0.229 ***	
<code>structural_composite</code> (C)	+0.454 ***	+0.021 ns	
<code>entropy_unified</code> ($\beta=1$)	+0.630 ***	+0.214 ***	
<code>entropy_adaptive</code>	+0.724 ***	+0.209 ***	
<code>total_combined</code>	+0.582 **	+0.209 *** (W/L 29/4)	
<code>entropy_target_ka</code>	+0.662 ***	+0.025 ns (clamp bug)	
<code>entropy_telgarsky</code>	+0.598 **	+0.040 ns	
<code>cross_layer_mi</code> (Path F, <code>ph11</code>)	+0.000 ns		—
<code>total_correlation_mi</code> (Path I, <code>ph12</code>)	–1.014 *** ($\lambda = 0.1$)		—
<code>total_correlation_mi</code> (Path I, <code>ph15</code>)	+0.306 . ($\lambda = 0.03$)		—
<code>entropy_lyapunov</code> (<code>ph18/18c</code>)	—		— — (FashionMNIST: highway-10 +0.092 * @ 10-seed)

Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, “ns” = not significant.

6 Where the manuscript stands

The defensible claim.

Across 90+ paired-seed experiments spanning 9 datasets and 5 architectures, the normalised-entropy formulation $L_{reg} = \lambda \cdot H(A) / \log_2 \text{rank}(A)$ strictly Pareto-dominates the unnormalised form: it preserves all significant positive effects, neutralises the four significant negative effects observed under the unnormalised regulariser, and admits a principled per-dataset H^ derived from the Kolmogorov-Arnold representation theorem.*

Negative result with a calibration law (Paths F & I).

The activation-side family (Path F `cross_layer_mi`, Path I `total_correlation_mi`) is dominated by Path B everywhere on the universality matrix. Path I's collapse at the conventional λ follows the calibration law $\lambda_{\text{multi}} \sim \lambda_{\text{scalar}}/L$, empirically verified at $L = 4$ (spirals collapse at $\lambda \geq 0.1$, productive band at $\lambda \approx 0.025\text{--}0.03$) and at $L \approx 41$ (resnet-20: rescaled $\lambda \approx 0.0005$ recovers the same null-positive band as Path B at $\lambda = 0.01$, not an additional lift). The single positive trend is `total_correlation_mi` at $\lambda = 0.03$ on spirals (+0.306 pp, $p \approx 0.08$, reproduced to three decimals across phases 12 and 15). Multi-term wrapping (`entropy_lyapunov`) on the PyTorch deep-architecture matrix yields two confirmed positives (`fashion_mnist_highway_10` at $L=10$, +0.092 pp, $p < 0.05$ at 10-seed; `mnist_resmlp_40_hymeko` at $L=40$, +0.213 pp, $p < 0.05$ at $\lambda_{a,b}=0.5$) once λ is rescaled per the calibration law for the deep variant. Highway-20 remains a genuine depth-fragility null across the full $\lambda \in \{0.5, 1.0, 2.0, 5.0\}$ grid.

Architectural-parity claim (phase 18b).

HyMeKo-emitted networks — compiled from `data/nn/*.hymeko` sources via the `torch_dataflow` back-end through Tier-2 `ResidualBlock` / `HighwayBlock` blocks — statistically mirror their hand-written PyTorch counterparts on the multi-term-wrapping matrix. The canonical IR therefore round-trips through to a runnable bench fixture without distorting regulariser dynamics.

Open work.

- Phase 7c (in progress): extend the universality stress to 5 architectures $\times$ 9 datasets.
- Phase 9 (queued): composability vs dropout / batch-norm / weight-decay — three new paper claims hinge on this.
- Re-run phase 13 once CapsMLP is refactored to expose its routing as a hookable submodule (the activation-hook [WARN] guard now prevents the silent-collapse failure mode that invalidated the original phase 13).
- Fix the ceiling clamp in `entropy_target_ka` from 1.0 to 0.75 (matches the H^* sweep collapse threshold).
- Reproducibility: all CSVs are seeded; full re-run is one overnight.

A File layout

`python/benches/thesis_iv_hard/run_benchmark.py`

main harness; defines the 12 arms, the model classes, and the DATASETS dictionary.

`python/benches/thesis_iv_hard/run_overnight_views_ph?.sh`

one shell script per phase; each is a sequence of RUN invocations of the harness.

`python/benches/thesis_iv_hard/aggregate_views_suite.py`

rolls all CSVs into RESULTS_VIEWS_SUITE.md.

`python/benches/thesis_iv_hard/aggregate_hstar_sweep.py`

H^* sweep aggregator; parses the phase-7 log.

`python/benches/thesis_iv_hard/test_sanchez_giraldo.py`

sanity test for the matrix-based Rényi-2 estimator.

`python/benches/thesis_iv_hard/compare_entropies.py`

weight-side vs activation-side entropy comparison study.

reports/sanchez_giraldo_framework.\{md,pdf\}

math draft for the activation-side framework (Path F unified with E, G, H).

reports/phase7c_brief.\{md,pdf\}

descriptions, expected outcomes, and exact bash commands for the 12 phase-7c runs.

reports/phase9_brief.\{md,pdf\}

purpose, configuration matrix, per-cell bash commands, and four paper-claim hypotheses with acceptance criteria for the 20 phase-9 paired comparisons.

reports/thesis_iv_views_suite.\{tex,pdf\}

earlier 5-page LaTeX report up to phase 7 (60-experiment summary).

RESULTS_VIEWS_SUITE.md

aggregated all-experiments table at repository root.